

Observation and mitigation of RF-plasma-induced damage to III-nitrides grown by molecular beam epitaxy

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Abstract – In this work, radio-frequency (RF) plasma-induced damage to III-nitrides surfaces and bulk defects are observed, and mitigated. It is shown that for InN films, the surface is more sensitive to plasma-induced damage than GaN films, as observed via atomic force microscopy (AFM) and reflection high energy electron diffraction (RHEED). In order to isolate any possible plasma-induced damage, a growth window for InN is established, and temperature ranges are determined for other damaging effects which include: roughening due to low adatom mobility, InN decomposition, and indium desorption. In-situ plasma monitoring and optimization are accomplished with a combination of optical emission spectroscopy (OES) as well as a remote Langmuir probe. It is shown that by increasing the plasma nitrogen flow, the positive ion content increases; however, the ion acceleration potential reduces. Additionally, a reduced RF plasma power results in a reduction of atomic nitrogen species. These plasma species and energetic variations result in variations in the bulk unintentional background electron concentrations observed by room temperature Hall Effect measurements of ~1 μm thick InN films. By increasing the nitrogen flow from 2.5 to 7.5 sccm for a constant RF power of 350 W, the background electron concentration decreases by 74% from $1.36 \times 10^{19} \text{ cm}^{-3}$ to $3.54 \times 10^{18} \text{ cm}^{-3}$, while maintaining a smooth surface morphology. Additionally, photoluminescence spectra indicate optical emission energies shift from ~0.81 to 0.71 eV (closer to the fundamental bandgap of InN) by limiting the damaging plasma species. Finally, conditions are presented to further minimize plasma-induced damage in III-nitride devices.

I. INTRODUCTION

The III-nitrides demonstrate a band gap as low as ~ 0.7 eV for InN ¹, a high light absorption coefficient², and a high radiation resistance³ which are all desirable for a variety of optoelectronic devices such as light emitting diodes (LEDs) and solar cells.

III-nitride devices are commercially available, such as blue LEDs due to the low indium content used in the $\text{In}_x\text{Ga}_{1-x}\text{N}/\text{GaN}$ quantum well active layers ($x < 0.15$), resulting in a low lattice-mismatch between the $\text{In}_x\text{Ga}_{1-x}\text{N}$ quantum well and GaN barrier and high electron-hole wave function overlap. Increasing the indium content in both the quantum wells and barriers can enable longer wavelength emission while minimizing the generation of additional lattice-mismatch induced defects in the active layer. Thus, high indium content $\text{In}_x\text{Ga}_{1-x}\text{N}$ with low defect densities, and single-phase crystallinity are essential to realize the next generation of long wavelength III-nitride multi-quantum well (MQW) LEDs.

In addition to LEDs, III-nitride solar cells have been demonstrated with active layers consisting of either $\text{In}_x\text{Ga}_{1-x}\text{N}/\text{GaN}$ MQWs^{4–6} or single semi-bulk $\text{In}_x\text{Ga}_{1-x}\text{N}$.^{7,8} In order to optimally design single junction and tandem $\text{In}_x\text{Ga}_{1-x}\text{N}$ solar cells for optimal utilization of the solar spectrum, $\text{In}_x\text{Ga}_{1-x}\text{N}$ absorbing layers with $x > 0.5$ is required.⁹ Presently, $\text{In}_x\text{Ga}_{1-x}\text{N}$ solar cells are limited in photovoltaic performance with efficiencies less than 4%¹⁰ primarily due to the limited indium content, and the poor crystal quality of the active layer and, thus, low carrier extraction efficiency. Therefore, improving the crystal quality of high indium content III-nitrides is essential for next generation LEDs, solar cells, and other optoelectronic devices.

III-nitride molecular beam epitaxy typically uses one of two nitrogen sources. In one form, ammonia is supplied to the growth surface and thermally “cracked” to provide active nitrogen to the crystal. Achieving high quality films via ammonia-based MBE becomes significantly more challenging at low growth temperatures due to the inefficient cracking of nitrogen.¹¹ The high growth temperatures in ammonia-based MBE required to growth high quality nitride-based films is not suitable for the growth of high indium content $\text{In}_x\text{Ga}_{1-x}\text{N}$ films. However, plasma-assisted molecular beam epitaxy (PAMBE) which utilizes a radio frequency (RF) plasma source to dissociate gaseous molecular nitrogen can be extended to colder substrate temperatures as low as 275 °C for InN ¹² and is thus capable of growing single phase $\text{In}_x\text{Ga}_{1-x}\text{N}$ for all x from $0 \leq x \leq 1$.^{13–15}

For III-nitride films, as the indium content increases, the band gap decreases, and, thus, the bond strength decreases. As the bond strength decreases, the film becomes more susceptible to plasma-induced damage. Crystal damage during the growth arises from impinging high kinetic energy particles which, in turn, result in enhanced decomposition, desorption of adatoms, and an increase in surface dangling bonds. The mentioned drawbacks can detrimentally reduce the surface diffusion length and may even prevent step-flow growth. Given the PAMBE nitrogen sources are remote sources with low energy ions, typically less than 6 eV of kinetic energy filtering out through a series of orifices, any detected ion damage would likely be worsened in direct plasma deposition techniques with high ion energies such as sputtering. Therefore, in

order to optimize III-nitride film quality and realize the next generation of optoelectronic devices, damage from the plasma discharge must be characterized and minimized.

In a PAMBE reactor, typically the plasma chamber is a pyrolytic boron nitride chamber in series with the nitrogen gas source and the growth chamber. The plasma chamber is excited by an RF electromagnetic source able to energetically excite ($< \sim 9$ eV) and even crack (> 9.7 eV) molecules of nitrogen producing various populations of excited atomic and molecular nitrogen along with both positive and negative ions and free electrons. When nitrogen flows through the plasma chamber, a conductance limitation enabled by an aperture plate at the plasma chamber outlet results in pressures in the order of millitorr. The RF energy excited gas leaks out into the growth chamber which is operated in the 10^{-5} - 10^{-6} Torr pressure range. Along with the excited neutral charged gas emission through the aperture plate, both electrons and ions escape and all species are directed in a molecular beam (essentially collisionless) toward the substrate. Since the vacuum acts as a dielectric capacitance and the substrate is typically grounded to the chamber potential, the relative differences in the positive to negative species (positive ions, negative ions, and electrons) emission rates induce a net “floating potential” in steady state between the chamber/substrate and the plasma itself. Direct bombardment of ions from inside the plasma to the substrate is limited by the aperture plate containing narrow holes, and so the primary source of acceleration energy for any ions towards the substrate is from the floating potential. The nature of the plasma discharge is determined primarily from the plasma bulb geometry, the size of each hole and conductance of the collection of holes in the aperture plate, the input nitrogen gas flow, and the applied plasma power.¹⁶ Specifically, the plasma pressure in the bulb is determined by the flow rate and the conductance of the aperture determines the mean free path for plasma collisions, and in large part, the acceleration potential. The geometry of the plasma bulb as well as the number of holes in the aperture plate modify the bulb volume and conductance.¹⁷ A wide range of applied plasma powers and nitrogen gas flows are used to study the optical and electrical characteristics of the plasma discharge and the crystal quality of the grown films.

The plasma discharge can be electrically characterized *in-situ* via a Langmuir probe.^{18,19} The current-voltage characteristics of the Langmuir probe can reveal the positive and negative ion contents and the ion acceleration potential, which is the voltage induced between the plasma and the substrate. Reducing the ion content, and minimizing the ion acceleration potential has been demonstrated to improve III-nitride film quality.²⁰

The optical behavior of the plasma can be characterized via optical emission spectroscopy (OES) through the rear plasma window port. OES provides a qualitative understanding of the molecular and atomic nitrogen species in the plasma. It is shown that the ratio of the molecular and atomic nitrogen species in the plasma changes by varying the input nitrogen flow and applied plasma power.²¹ The optical characteristics of the plasma measured by OES are weighted by the transmission properties of the rear plasma window and the PBN bulb.

OES measures both excited molecular and atomic nitrogen species, but cannot distinguish between ionic and neutral excited species. In contrast, a Langmuir probe can measure the charged particles in the form of ions and electrons, but cannot distinguish between molecular and atomic ions. Plasma characterization is best performed when combining these two techniques while recognizing their individual limitations.

In this work, InN, possessing the weakest molecular bond strength (7.72 eV/bond in the bulk) of the III-nitrides (compared to 9.12 and 11.52 eV/bond of GaN and AlN, in the bulk, respectively), is utilized as a sensitive material to characterize and optimize the plasma discharge.²² It is important to note that the surface bond energies may be significantly less than those of the bulk due to local near-surface crystalline softening and chemical catalytic energy additions. Thus, the plasma-induced damage in InN films needs to be isolated from other potentially damaging effects which can also generate defects in the film. These effects primarily include roughening due to low adatom mobility, InN decomposition, and indium desorption.

The Langmuir probe, OES, and InN crystal quality data are combined to provide insight and guidance towards optimal plasma conditions which minimize plasma damage of high indium content III-nitrides films.

II. EXPERIMENTAL PROCEDURE

In this study, a Riber 32 PAMBE reactor is employed to study the nitrogen plasma discharge and grow nitride-based films. GaN templates on sapphire were chosen as substrates. The templates were first backside metallized with 2 μm of tantalum to ensure temperature uniformity during the growth. Next, the templates were cleaned with a two-stage piranha clean consisting of 3:1 of $\text{H}_2\text{SO}_4\text{:H}_2\text{O}_2$ followed by a DI water rinse. The substrates were then loaded into an introductory chamber and outgassed at 200 $^\circ\text{C}$ before loading into the growth reactor. Inside the growth reactor, indium and gallium fluxes are supplied via a standard effusion cell and a Veeco dual-filament SUMO cell, respectively. A Veeco UNI-Bulb RF nitrogen plasma source was used to supply various nitrogen flows and powers ranging from 2.5 to 12.5 sccm and 350 to 550 W, respectively. The plasma source was equipped with an aperture plate containing ~2000 holes drilled with a #80 drill bit (0.0135" diameter). This is roughly half the total conductance used in our previous work that resulted in GaN growth rates of 9.8 $\mu\text{m}/\text{hour}$ ¹⁷. The plasma optical emission spectrum was monitored *in-situ* via an Ocean Optics HR-2000+ES spectrometer through both a rear window and the pyrolytic boron nitride (PBN) bulb. The flux gauge collector terminal is repurposed as a Langmuir probe and placed in a position that matches the film growth position, in order to accurately characterize the plasma in the growth environment. A Keithley 6517A electrometer is connected to the collector pin feedthrough to sweep the voltage between -100 and 100 V and sense current. Growths were achieved by utilizing the metal modulated epitaxy (MME) growth technique, presented in our previous works.^{12,23} Reflection high energy electron diffraction (RHEED) was used to monitor the MME growths *in-situ*.

III. RESULTS AND DISCUSSION

The sensitivity of nitride-based surfaces to the plasma discharge is first studied by comparing the surface quality of GaN and InN films which were exposed to the plasma discharge post-growth with a nitrogen plasma input flow of 2.5 sccm and an applied RF power of 350 W. Figure 1 compares AFM and RHEED images of GaN and InN surfaces before and after exposure to the plasma discharge for 5 minutes at growth temperatures of 600 and 300 °C for the GaN and InN films, respectively. The GaN surface shows a negligible difference in morphology when exposed to the post-growth plasma discharge, whereas the InN surface exhibits significant texturing. It should be noted that exposure of the plasma to the GaN surface for even a few minutes resulted in a rising specular RHEED spot intensity associated with slight surface roughening, as seen in Figure 1. Longer post-growth plasma exposure (e.g., one hour) of GaN resulted in a significant surface roughening.

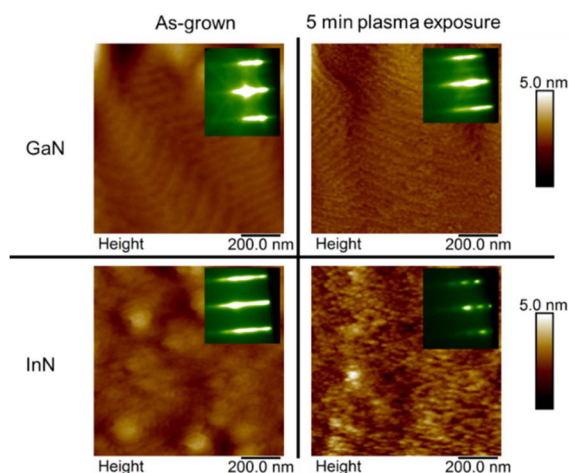


Figure 1. AFM images of as-grown and post-growth plasma exposed GaN and InN films. The InN surface is degraded by plasma exposure (inset: associated RHEED images.)

InN film quality can be deteriorated due to a variety of thermal related factors such as high growth temperatures resulting in indium desorption and/or InN decomposition and low growth temperatures resulting in low adatom mobility. In order to isolate the effects of the plasma damage on the InN film quality, a temperature growth window, free of the mentioned thermal related effects, was first established. Figure 2 illustrates the onset of each thermal related effect and their associated RHEED patterns. First, at low substrate temperatures (e.g., 225 °C), the indium adatom mobility is low, which results in the surface roughening and spotty RHEED patterns. Next, when the substrate temperature is set to greater than 250 °C (e.g., 300 °C) the film exhibits a 1 x 1 RHEED streaky pattern and, thus, a smoother InN surface. Finally, at higher growth temperatures (e.g., 450 °C), the RHEED patterns dim due to the accumulation of excess indium metal on the surface, which scatters/blocks the electron beams as a result of InN decomposition.

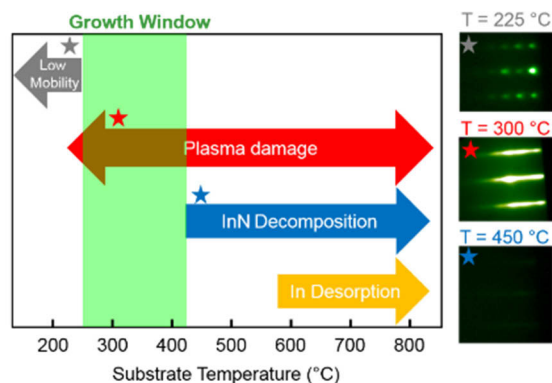


Figure 2. Substrate growth temperature window for MME InN for isolation and characterization of the plasma damage effects. The RHEED images indicate a rough surface for substrate temperatures that are low and InN decomposition for substrate temperatures that are high.

The onset of InN decomposition and indium desorption were analyzed to determine a more accurate temperature boundary and extract their corresponding activation energies. First, InN decomposition was studied by growing InN via MME with a growth temperature of 360 to 430 °C. When InN decomposes into indium and nitrogen species, the indium metal remains on the surface and contributes to the total metal dose as long as the growth temperature is lower than the temperatures at which indium desorbs from the surface. For a constant plasma condition and, thus, nitrogen flux, the excess indium introduced from the decomposition of the underlying InN film can be quantified by measuring the additional time required to consume the total metal dose when compared to lower growth temperatures less than 360 °C without indium decomposition. RHEED transients show that the InN decomposition rate increases by increasing the growth temperature, leading to a longer metal consumption times (see Figure 3). As seen in Figure 3, by increasing the substrate temperature, the duration of the highlighted region, associated with the decomposed In metal from the underlying InN layers, increases.

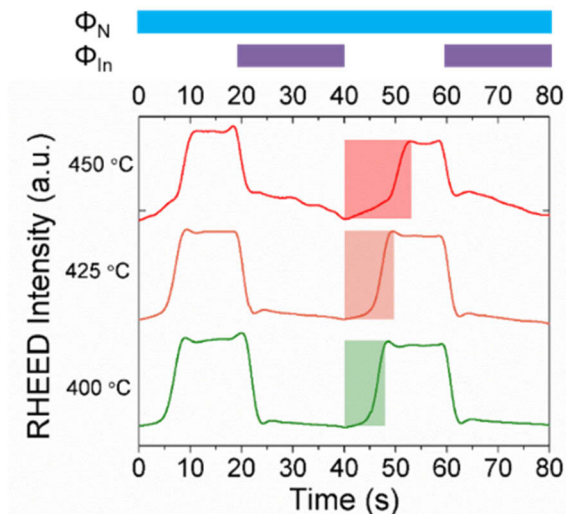


Figure 3. MME RHEED intensity transients during the growth of InN at different substrate temperatures where InN decomposition occurs. At higher growth temperatures, InN decomposition contributes additional indium metal to the surface.

The increased amount of time required to consume the indium metal on the surface can be converted into an excess flux of indium in atoms/cm². Figure 4 illustrates the exponential nature of the decomposition flux vs substrate temperature. The extracted InN decomposition activation energy is ~ 1.77 eV which is close to the previously reported value of 1.78 eV.²⁴

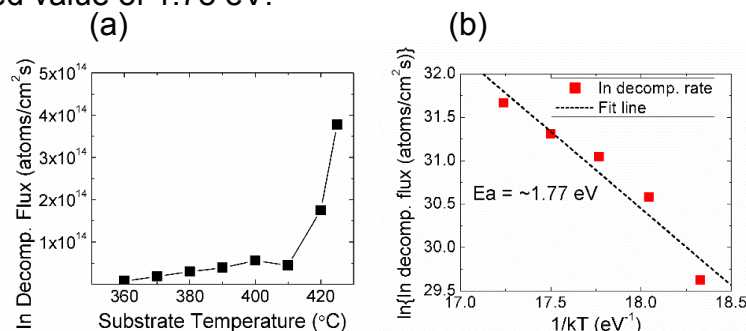


Figure 4. InN decomposition flux in the form of excess In flux (atoms/cm²) present on the surface at various substrate temperatures for both (a) linear and (b) Arrhenius plots.

In addition to the InN decomposition, the substrate temperatures in which indium desorption occurs are identified. Unfortunately, it is difficult to directly measure the indium desorption from an InN surface by RHEED transients due to the InN decomposing at significantly lower substrate temperatures than those required for desorption, as depicted in Figure 2. Thus, at first, an indium dose of 5.1×10^{16} atoms/cm² is supplied to a GaN surface without any supplied nitrogen flux. Subsequently, most of the desorbed indium is indium desorption from the metallic indium surface, and not directly measuring desorption from the GaN surface. Indium desorbing from indium on a GaN surface should provide a rough estimate of the substrate temperature range where significant indium desorption occurs. Figure 5 details RHEED intensity transients where the indium dose is evaporated onto the GaN

surface in the first 20 s and then desorbed. As seen, the RHEED intensity falls as the indium accumulates on the surface and rises to its initial value once it is completely desorbed. In addition, Figure 5 illustrates that the same amount of indium desorbs faster at higher substrate temperatures.

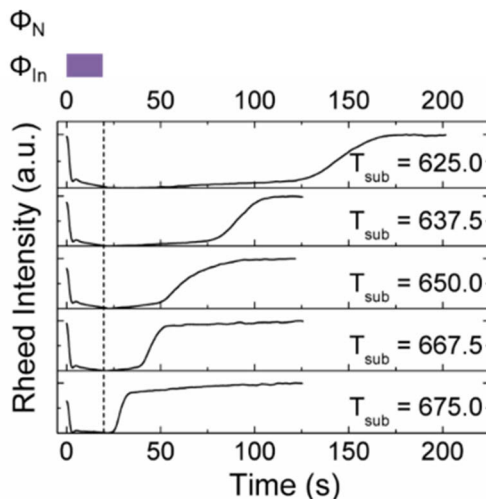


Figure 5. RHEED intensity transients after evaporating 5.1×10^{16} atoms/cm² indium onto a GaN surface for 20 s. RHEED transients illustrate the increasing rates of indium desorption with increasing substrate temperature.

The time to desorb the indium dose on the surface can be then plotted vs substrate temperature (see Figure 6). As seen in Fig. 6, the desorption flux exponentially increases with increasing substrate temperature. The Arrhenius plot indicates an indium desorption activation energy of ~ 2.32 eV, which is most likely indium desorbing from the metallic indium but does closely match with Gallinat *et al.* reported value of 2.49 eV.²⁴

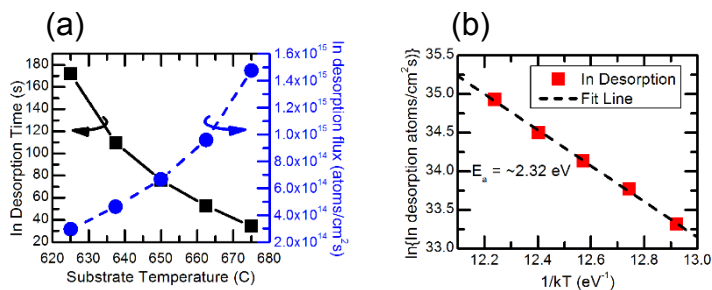


Figure 6. (a) In desorption time and associated desorption flux and (b) Arrhenius plot of the desorption flux vs substrate temperature.

By *in-situ* analysis of RHEED patterns and intensity transients, the substrate temperature boundaries for the onset of surface roughening due to low adatom mobility, InN decomposition, and In desorption are established. These temperature boundaries define a growth window for InN containing devices in the range of 250-400 °C. In this work, InN films grown in this window act as a sensitive material to isolate any potential influence of the plasma conditions on the crystal quality.

The plasma can be electrically characterized *in-situ* via a flux ion gauge collector pin repurposed into a Langmuir probe. A Keithley 6517A electrometer was connected externally to the collector pin feedthrough. By placing the manipulator into the flux measurement position, the Langmuir probe is directed towards the plasma discharge. When sweeping voltage and measuring current, any ions/electrons collected are measured as a contribution to the total current.

The plasma is then electrically characterized *in-situ* via the flux ion gauge collector pin repurposed into a Langmuir probe and placed in the growth position. By sweeping voltage and measuring current, any ion/electrons contribute to the total current. At a high reverse (forward) bias, most of the impinging negative electrons (positive ions) are repelled, and mainly the positive ions (electrons) are collected, resulting in a measured positive (negative) current. Since, in this setup, the Langmuir probe is outside the plasma bulb, and the density of the ions and electrons are often uneven, the probe may not collect an equal contribution of positive ions and negative electrons at zero applied bias. Instead, the plasma bulb will retain a net charge (+Q) that results from excess charge (−Q) ejected from the plasma towards the probe which establishes a steady state acceleration voltage (V_a) according to $Q = CV_a$ (where C is the combined series capacitance of (a) the internal plasma and (b) the vacuum dielectric capacitance). The internal plasma capacitance depends mainly on the plasma chemistry modified by the plasma input conditions and bulb dimensions. The vacuum dielectric capacitance is mainly associated with the aperture to chamber (or sample) location. There corresponds to the series capacitances, a plasma potential (V_p), and a floating potential (V_f). The Langmuir probe bias that cancels the net charge expelled from the plasma and, thus, results in a zero measured current can be identified as V_f . Generally, V_p inside the plasma bulb (internal plasma) is found by the inflection point of the Langmuir probe I-V data in the positive potential region.²⁵ The difference between V_p and V_f is defined as the ion acceleration potential (V_a). Thus, a Langmuir probe can provide an understanding of the plasma contents, along with the induced ion acceleration potential between the plasma and the sample substrate.

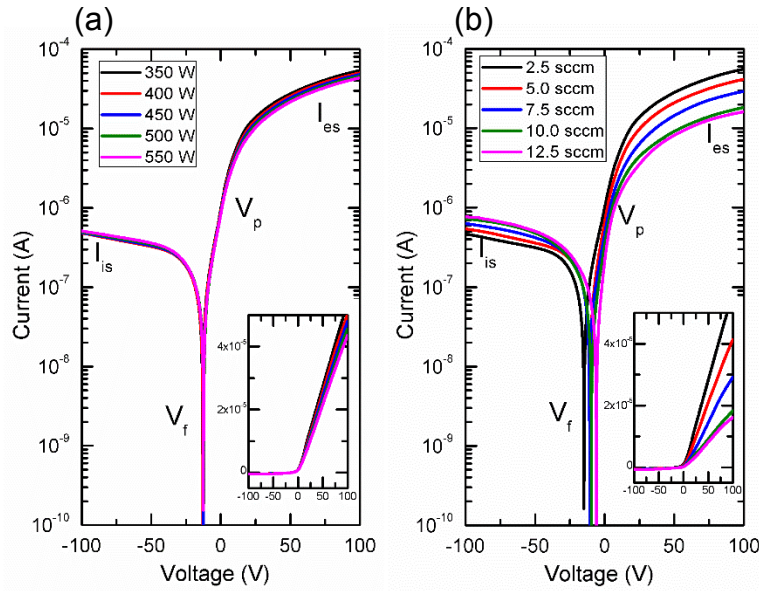


Figure 7. Semi-log and linear current-voltage characteristics of the Langmuir probe for different plasma conditions: (a) 350 to 550 W RF applied power with a constant 2.5 sccm N_2 flow and (b) 2.5 to 12.5 sccm N_2 flow with a constant 350 W RF applied power. The ion saturation current (I_{is}), electron saturation current (I_{es}), floating potential (V_f), and plasma potential (V_p), are all labeled.

The Langmuir probe's current-voltage characteristics of the plasma at various power and flow conditions are presented in Figure 7. The positive ion saturation current (I_{is}) and negative electron saturation current (I_{es}), the zero-current floating potential (V_f), and the plasma potential (V_p) are all labeled. It is evident that the plasma related parameters (including I_{es} , I_{is} , and V_a) do not change with increasing plasma power, as the plasma density is proportional to the square root of power. Contrarily, as seen in Figure 7(b), an increased nitrogen gas flow results in an increased positive ion content, a reduced negative ion/electron content, and, thus, a more neutral plasma discharge which lowers the floating potential.

In addition to the floating potential, the plasma potential must be found to determine the overall acceleration potential ($V_a = V_p - V_f$). The plasma potential is best observed when the inflection point is easily identifiable in the current-voltage characteristic. Hershowitz details the best configuration for observing this inflection point when employing a semi-infinite planar probe inside the plasma.²⁶ Unlike Hershowitz, in this paper, a circular flux gauge collector pin is utilized as a Langmuir probe outside the plasma, and, thus, the current-voltage inflection point and the plasma potential are difficult to extract, as reported by Azooz.²⁷ However, some of the plasma potential values can be extracted by data smoothing and numerical differentiation of the I-V characteristics, as seen in Figure 8.

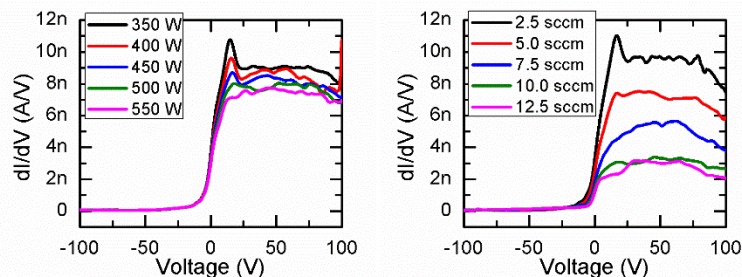


Figure 8. A numerical derivative of the Langmuir probe data for various (a) plasma powers and (b) nitrogen flows.

The derivative data shown in Figure 8(a), which correspond to the various RF powers and nitrogen flows, have identifiable peaks, associated with the plasma potentials. The floating potential, extractable plasma potentials, and acceleration potential ($V_p - V_f$) are plotted in Figure 9.

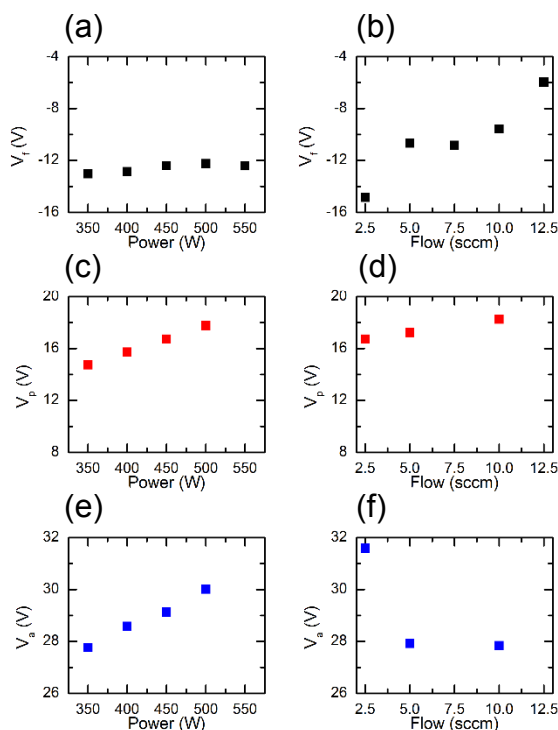


Figure 9. Plasma parameters extracted from the Langmuir probe current-voltage characteristics: (a-b) floating potential, (c-d) plasma potentials, and (e-f) acceleration potentials for various RF powers and nitrogen flows.

As seen in Figure 9, the magnitude of V_f is significantly reduced (by approximately 8 V) when increasing the nitrogen flow from 2.5 to 12.5 sccm, however, it negligibly changes by increasing the RF power from 350 to 550 W. In addition, the V_p increases by increasing the power and flow, which results in increased plasma density. The increase in the plasma density can be seen in Figure 10, as V_p linearly increases with the square root of the RF power.

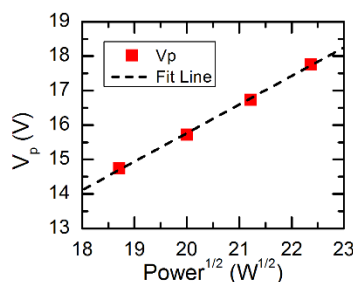


Figure 10. Plasma potential vs the square root of the RF power and the associated linear fit.

The combined effects of V_p and V_f result in a rise of V_a by ~ 1.4 V for an increase in the power from 350 to 500 W and a fall of ~ 3.8 V for an increase in the nitrogen flow from 2.5 to 10.0 sccm.

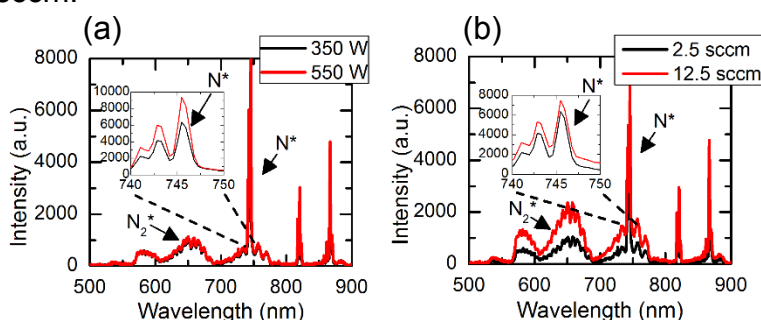


Figure 11. Optical emission spectroscopy for different conditions: (a) 350 and 550 W RF applied power with a constant 2.5 sccm N_2 flow, and (b) 2.5 and 12.5 sccm N_2 flow with a constant 350 W RF applied power (inset: a narrow wavelength range (740-750 nm) to view excited atomic nitrogen peaks).

The excited molecular (N_2^*) and atomic nitrogen (N^*) contents of the plasma are extracted from the OES data. The broad and low intensity peaks, displayed in Figure 11(a) and Figure 11(b), are excited molecular nitrogen, while the sharp and narrow peaks are associated with excited atomic nitrogen.^{16,21} The spectrum lines of 582 and 745 nm are chosen as representatives of N_2^* and N^* , respectively. Figure 12 displays atomic and molecular peak intensities vs plasma power and flow. Interestingly, the atomic peak intensity monotonically increases with increasing power, as seen in Figure 12(a), but saturates at high flows (see Figure 12(b)). Additionally, the molecular peak intensity has little correlation with increasing power (see Figure 12(c)), but monotonically increases with the increasing flow (see Figure 12(d)). While the excited atomic nitrogen peak intensity is much higher than the molecular nitrogen in Figure 11 and Figure 12, the integral width of the spectra is dominated by the molecular species suggesting that the atomic nitrogen percentage of the plasma is relatively small.

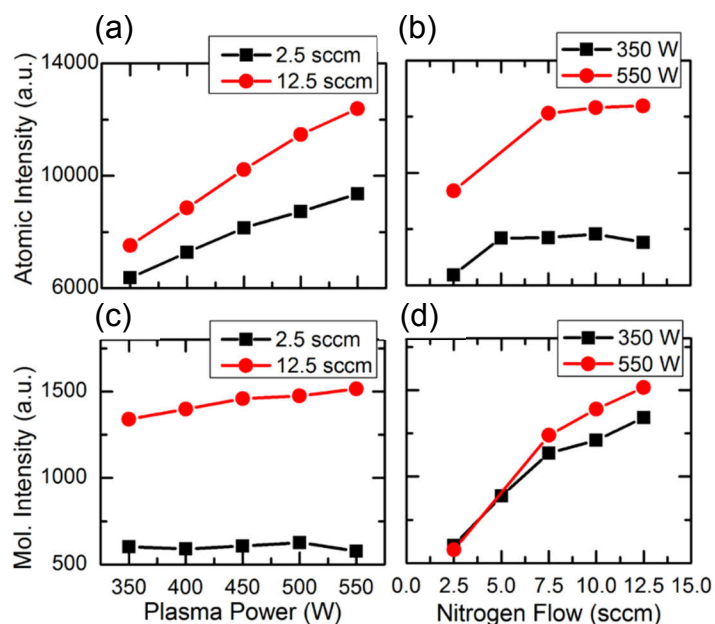


Figure 12. Peak OES intensity data for various plasma conditions at (a-b) atomic 745 nm emission and (c-d) molecular 582 nm emission.

By comparing the OES and Langmuir probe data, one can conclude that by raising the plasma RF power, the significant rise in N^* only results in a slight increase of positive ion content and V_a . The negligible difference in Langmuir probe data would imply that the percentage of the total plasma density and the ion content of the plasma are minimally affected by the atomic nitrogen content and this is consistent with the growth rate data which will be presented later. As will be also shown later, even though the atomic nitrogen content does not dominate the plasma characteristics, it correlates to the defect concentration of the films and, likely, is the source of surface roughening, observed in Figure 1, and the unintentional background electron concentrations of InN films.

Next, the effects of increasing the plasma nitrogen flow on the OES and Langmuir probe data are compared. The large increase in excited molecular nitrogen content from an increase in flow results in a larger positive ion content and reduced ion acceleration potential. At first, the increase in positive ion content with the increased flow may appear disconcerting. However, it is accompanied by a significant decrease in the acceleration potential responsible for the kinetic energy of ionized species that potentially causes degradation of the film quality. This is not surprising as the increased flow results in increased plasma bulb pressure and, thus, a decrease in the mean free path, described by Eq. (1), where the mean free path (l) is calculated from the Boltzmann constant (k), the temperature (T), the pressure (p), and the molecular diameter (d_m^2). This equation dictates that the smaller atomic nitrogen will have a longer mean free path than molecular nitrogen, thus, traveling longer before being scattered, leading to higher kinetic velocities.

$$l = \frac{kT}{\sqrt{2}\pi p d_m^2}, \quad (1)$$

The drop in the mean free path with higher flow results in increased scattering within the plasma bulb, limiting the kinetic energy of species and consequently lowering the floating potential of the remote plasma (over 8 V for an increase in flow from 2.5 to 10.0 sccm). The conductance of the aperture of the plasma bulb sets operational limits over which the plasma can be sustained. For the aperture used in this study, the pressure in the plasma bulb ranges from 1.5 to 7.4 mT and, consequently, a molecular nitrogen mean free path of 31 to 6 mm for a flow rate of 2.5 to 12.5 sccm. The atomic nitrogen mean free path for the same flow range, assuming independent gas species (for comparison purposes only), varies from 1,400 to 270 mm. Below the lower pressure limit, the plasma density is insufficiently high to support a plasma. At these lower flows, the acceleration potential is substantial due to the longer mean free path and, thus, higher obtainable scattering limited kinetic energies. At higher plasma pressures, the plasma mean free path is insufficiently long to allow energy transfer, thus, extinguishing the plasma. However, at the upper end of the allowable flow, the floating and acceleration potentials are substantially lower resulting in (as will be shown below) minor film damage. In a similar way, as the electrical conductance of the path from the aperture of the plasma to the sample (chamber ground) is increased by increased flow, the floating potential drops significantly due to the inability to support large voltages across a highly conductive remote plasma glow.

After optical and electrical characterization of the plasma, InN films are grown at various plasma conditions. Seven ~1 μm -thick InN films were grown via MME on GaN templates to determine the effect of plasma power and nitrogen flow on the material quality. First, three films were grown with a nitrogen flow of 2.5 sccm and varied plasma powers of 350, 450, and 550 W. Next, four additional films were grown with a standard plasma power of 350 W and nitrogen flows of 3.75, 5.0, 6.25, and 7.5 sccm. The InN films' surface morphology, X-ray diffraction, background electron concentrations, and photoluminescence are compared to elucidate the origin of the plasma damage, and predict optimal growth conditions. X-ray diffraction (not shown here) resulted in the (0002) omega rocking curve full width at half maximum (FWHM) values between 137 and 167" for all films. In addition, the (10 $\bar{1}$ 5) omega rocking curve FWHM values are between 316 and 337" for all films. A 5 x 5 μm^2 AFM image of the sample with the highest growth rate (grown with 350 W RF plasma power, 7.5 sccm N₂ flow and a corresponding growth rate of ~4.6 $\mu\text{m/hr}$) is shown in Figure 13. The film shows a RMS roughness of ~0.9 nm and a smooth spiral-hillock surface morphology, which is observed for all the grown InN films. Notably, the MME growth method with high growth rates yields high-quality InN films.

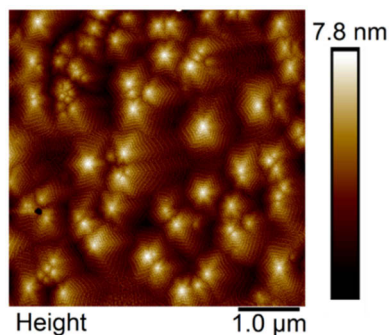


Figure 13. A $5 \times 5 \mu\text{m}^2$ AFM image of an InN film grown at $4.6 \mu\text{m/hr}$ exhibiting a 0.9 nm RMS roughness.

Since both ion damage and background contamination can affect the unintentional electron concentration, the films' growth rate and background electron concentration are compared, as shown in Figure 14, for both power and flow rate variations. The plasma condition of 350 W and 2.5 sccm nitrogen flow resulted in a growth rate of $2.01 \mu\text{m/hr}$ and a background electron concentration of $1.36 \times 10^{19} \text{ cm}^{-3}$. Growing an InN film with a plasma power of 550 W with 2.5 sccm nitrogen flow increased the growth rate up to $2.54 \mu\text{m/hr}$, and the background electron concentration escalated to $1.49 \times 10^{19} \text{ cm}^{-3}$. Interestingly, when growing a film with a nitrogen flow as high as 7.5 sccm, the resulting growth rate was $4.58 \mu\text{m/hr}$, and the background electron concentration reduced significantly to $3.54 \times 10^{18} \text{ cm}^{-3}$, unlike the trend for increasing plasma power. As a) both the residual background electron concentration and b) the change in electron concentration as a function of flow and, thus, growth rate are higher than expected for contamination, it is likely that the residual electron concentration is controlled predominantly by the ion damage. For example, when the growth rate increases from 2.01 to $4.58 \mu\text{m/hr}$ (by a factor of 2.28), the residual electron concentration drops from $1.36 \times 10^{19} \text{ cm}^{-3}$ to $3.54 \times 10^{18} \text{ cm}^{-3}$ (by a factor of 3.84). The reduction in the unintentional electron concentration suggests that the defect density in InN is controlled largely by the ion damage.

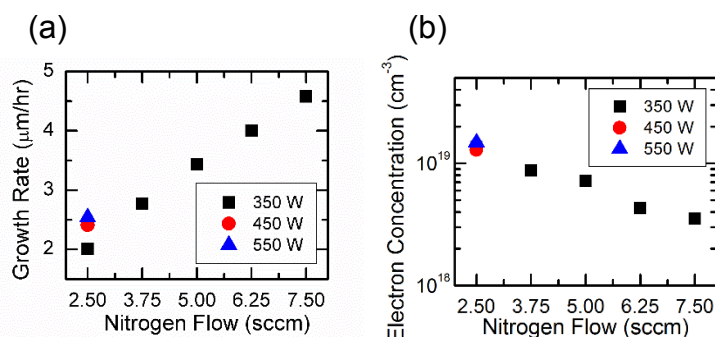


Figure 14. Comparison of (a) growth rates and (b) background electron concentrations for InN films grown at various nitrogen flow rates and plasma powers.

The increase in background electron concentration with increasing power may be attributed to the increase in atomic nitrogen species, as evident in Figure 12(a), with a

negligible increase in the molecular content, as shown in Figure 12(c). The ion acceleration potential in Figure 9(e) is also slightly increased with increased power. Thus, it is speculated that a larger ratio of atomic nitrogen to molecular nitrogen as well as an increased acceleration potential can cause damage to the crystal which results in a higher background electron concentration, similar to previous work for GaN¹⁶.

Photoluminescence (PL) spectra at low temperature (4 K) were taken for all seven InN films, as seen in Figure 15. The low temperature spectrums seem to be made up of two components: one invariant to changes in electron concentration and one dependent on electron concentration. The films' spectra show an emission at 0.77 eV which is invariant to electron concentration, well above the fundamental bandgap energy of $\sim 0.65\text{--}0.7$ eV for InN.^{28–30} The 0.77 eV peak is also present in room temperature PL (not shown). While a future, more in-depth PL study is planned, it is presently speculated that the 0.77 eV emission is associated with the surface electron accumulation layer due to the commonly reported fermi-level pinning observed in InN.³¹ At low temperatures, a second peak is also observed that varies with electron concentration. This peak is likely the band-to-band, or possibly the previously reported band-to-acceptor transition^{32,33} and can be seen varying from ~ 0.81 to 0.71 eV for electron concentrations varied from 1.49×10^{19} to $3.54 \times 10^{18} \text{ cm}^{-3}$. The ~ 100 meV shift from the highest to lowest electron concentration is likely due to the Moss-Burstein effect.³⁴ The smallest electron concentration samples' PL peak location approaches the fundamental bandgap. Thus, plasma optimization is essential for InN films, as observed both electrically and optically.

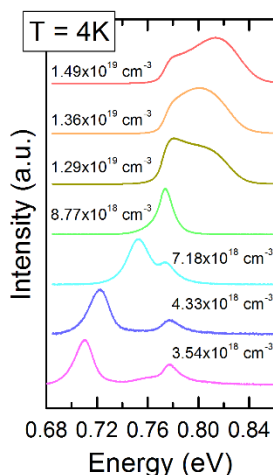


Figure 15. Photoluminescence spectra for $\sim 1 \mu\text{m}$ thick InN films grown under various plasma conditions at 4 K.

In order to minimize the plasma damage in high In content films, one must minimize the atomic nitrogen species per unit thickness (or growth rate) and minimize the ion acceleration potential. Ptak *et al.* demonstrated that the molecular nitrogen species are the species that control the growth rate and the atomic nitrogen species were speculated to generate point defects in the crystal.¹⁶ Thus, one should maximize the molecular to atomic nitrogen ratio and also minimize the ion acceleration potential in the plasma to minimize the plasma damage per unit thickness. Ptak *et al.* showed that

the size of the holes in the aperture plate could be used to adjust the atomic to molecular species but did not show the effect on the acceleration potential.

By plotting the OES molecular to atomic peak intensity ratio, one can determine the plasma condition necessary to maximize the molecular to atomic content and, thus, minimize the plasma damage. Figure 16 illustrates that at high flow and low power, the maximum ratio of molecular to atomic nitrogen is achieved, in agreement with the experimental results. This condition also results in the lowest floating potential and consequently the lowest acceleration potential as evident in Figure 9(b). Thus, minimizing the plasma damage for III-nitride films is achievable at low RF power and high nitrogen flow, while still maintaining smooth morphology.

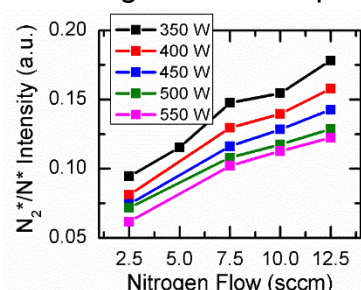


Figure 16. OES molecular to atomic peak intensity ratio for various plasma nitrogen flow rates and applied RF power.

IV. CONCLUSIONS

Plasma-induced damage has been observed in III-nitride films and has been isolated from other effects such as roughening from low adatom mobility, InN decomposition, and indium desorption. By utilizing a flux gauge as a Langmuir probe in combination with OES, the RF plasma in the PAMBE reactor can be characterized. It is determined that atomic nitrogen and a plasma induced acceleration potential result in damage to InN films as observed via an elevated background electron concentrations. A low RF applied power and a high nitrogen flow enable a high molecular to atomic nitrogen ratio as well as a high growth rate and minimal acceleration potential which is the optimal plasma condition for continuous growth of high indium content III-nitrides, necessary for future III-nitride optoelectronics.

ACKNOWLEDGEMENTS

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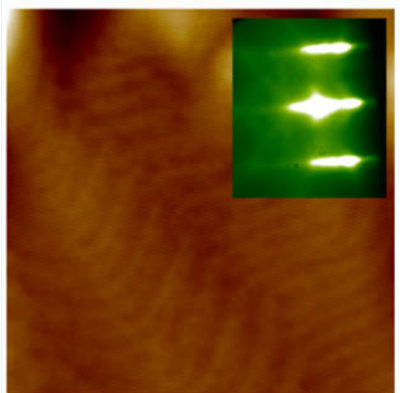
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As-grown

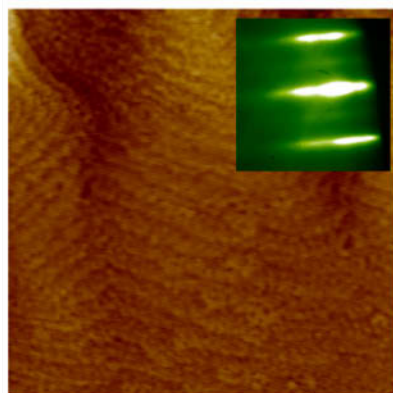
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GaN



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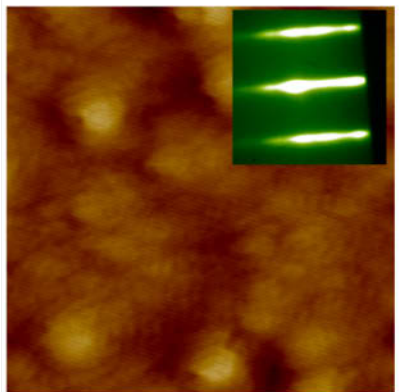
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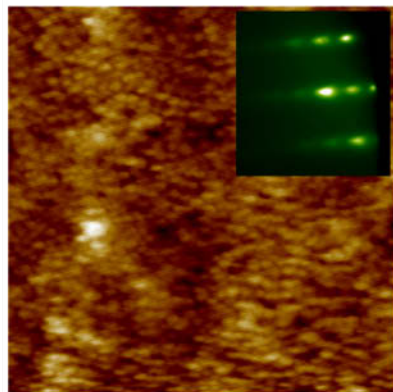


InN



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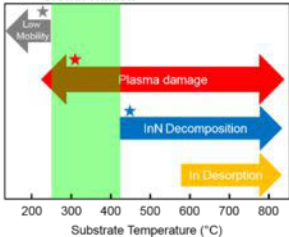
Height

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5.0 nm



Growth Window



T = 225 °C



T = 300 °C



T = 450 °C



Φ_N Φ_{In}

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RHEED Intensity (a.u.)

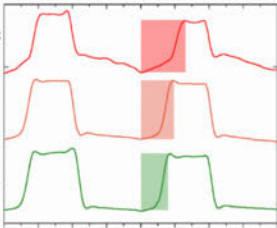
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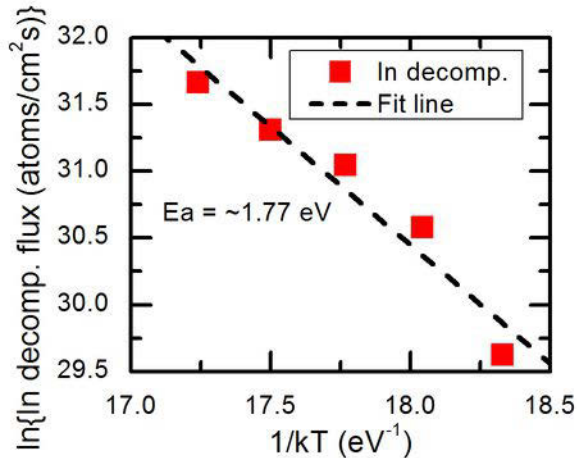
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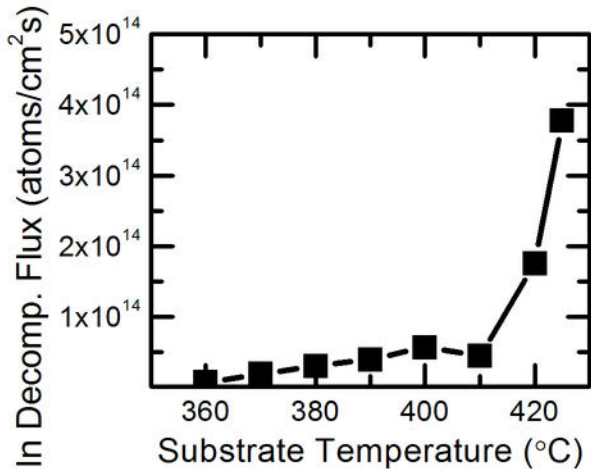
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Time (s)

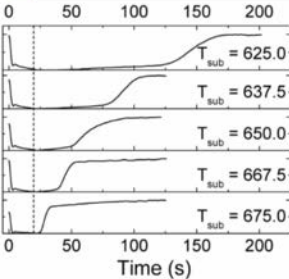


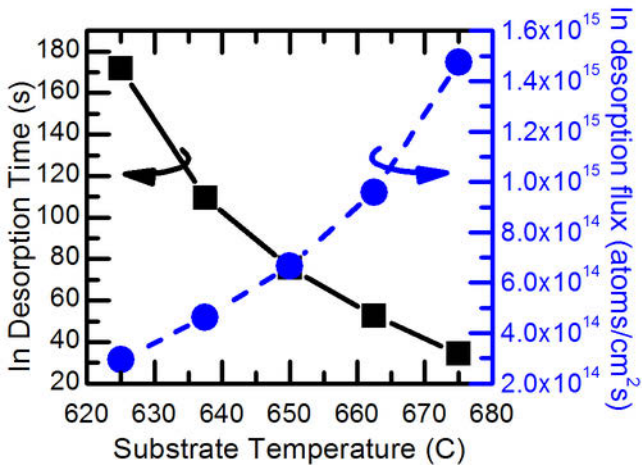


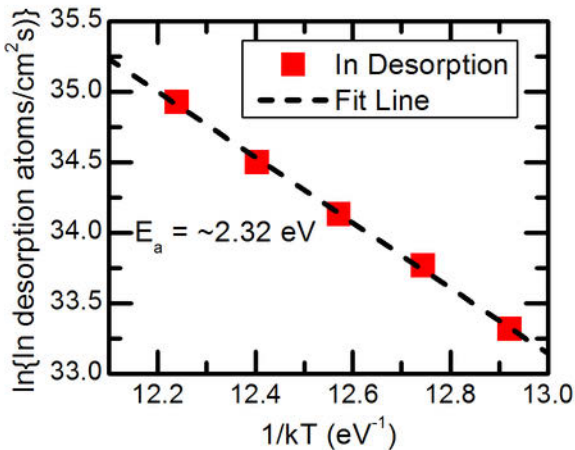


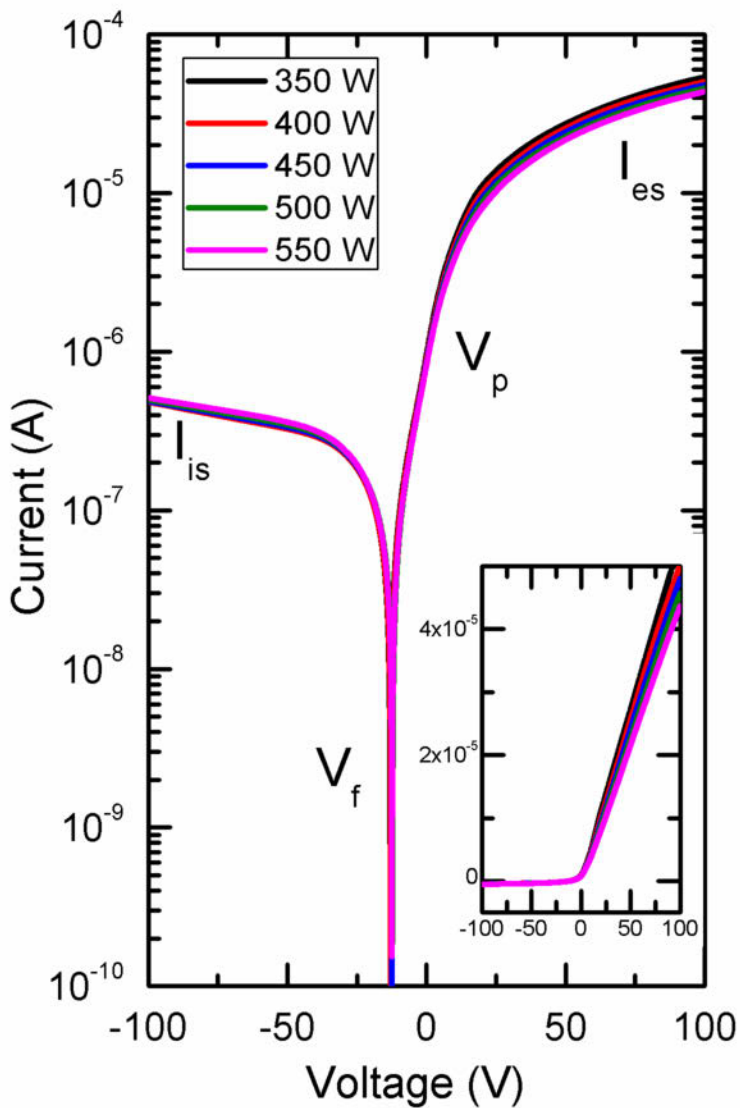
θ_N θ_{in} 

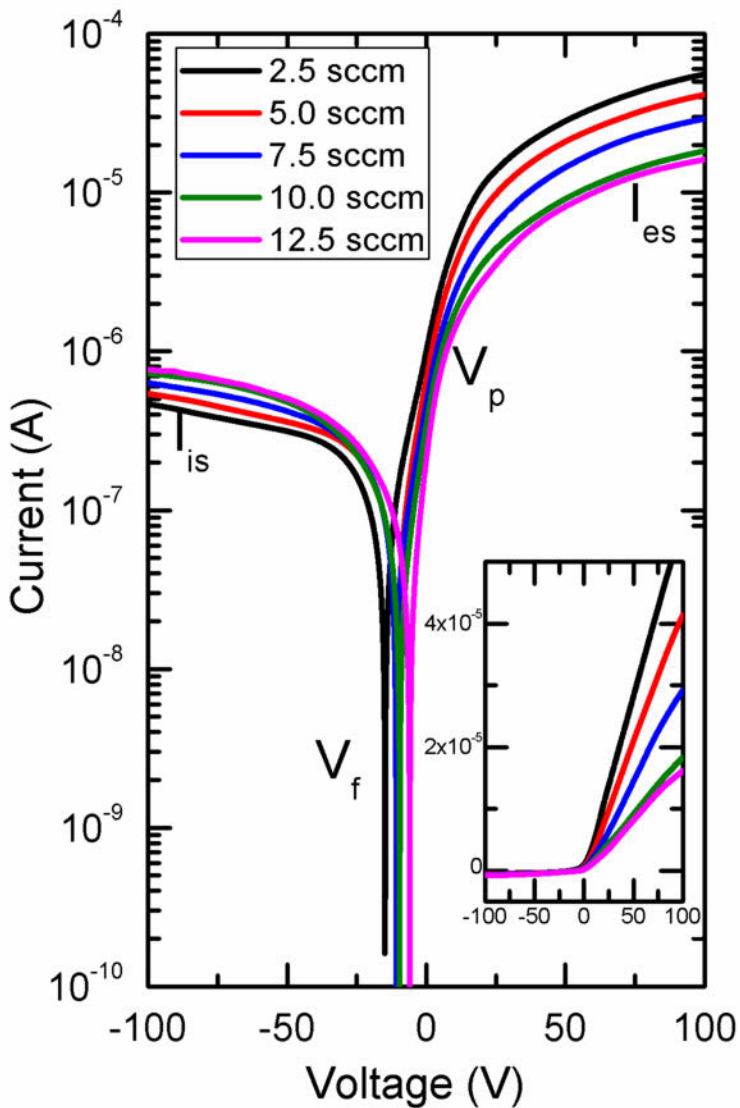
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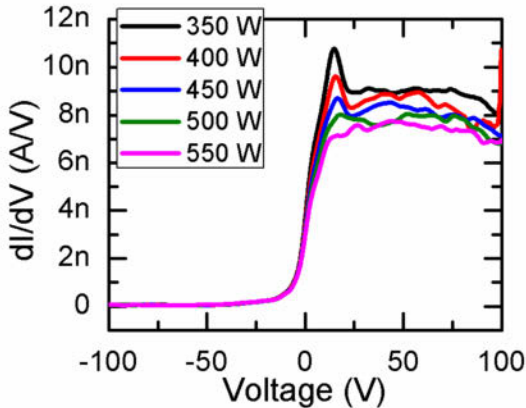


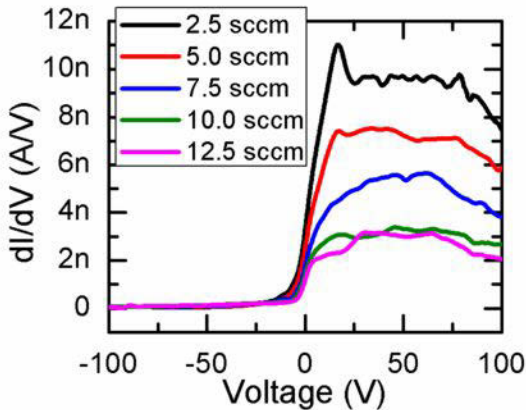


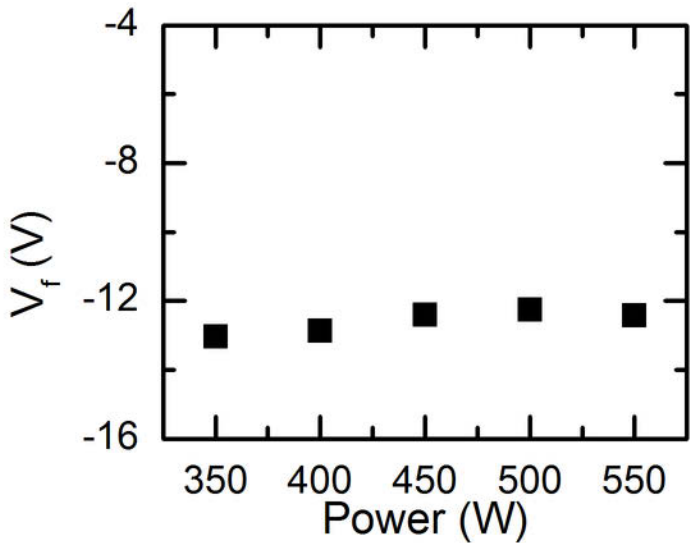


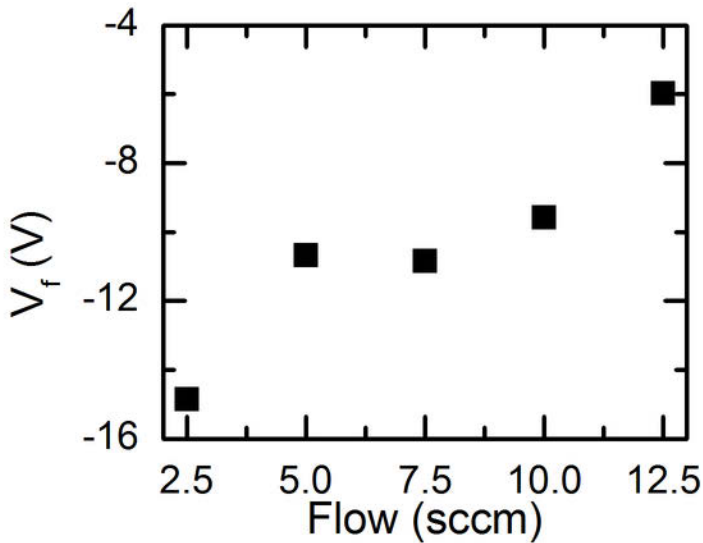


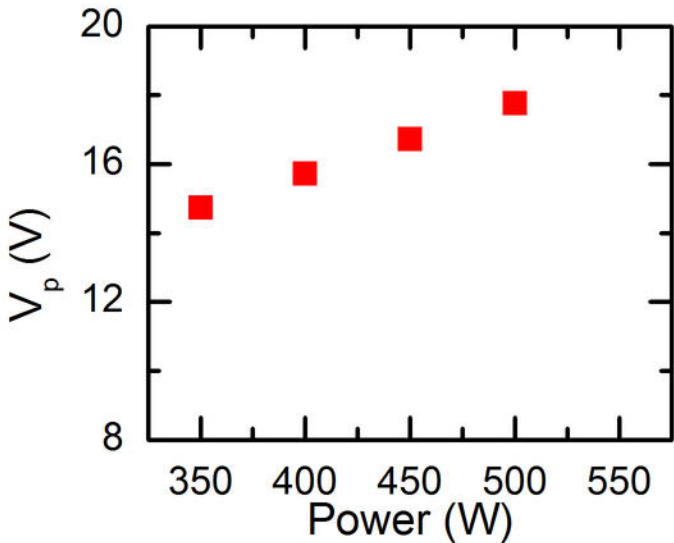


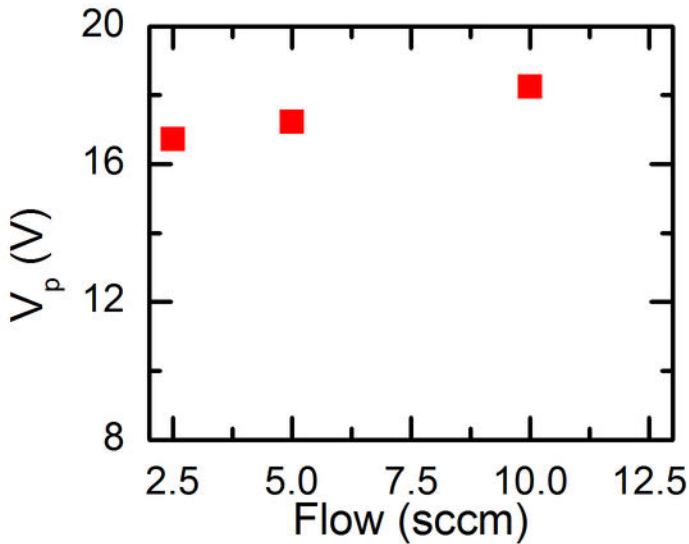


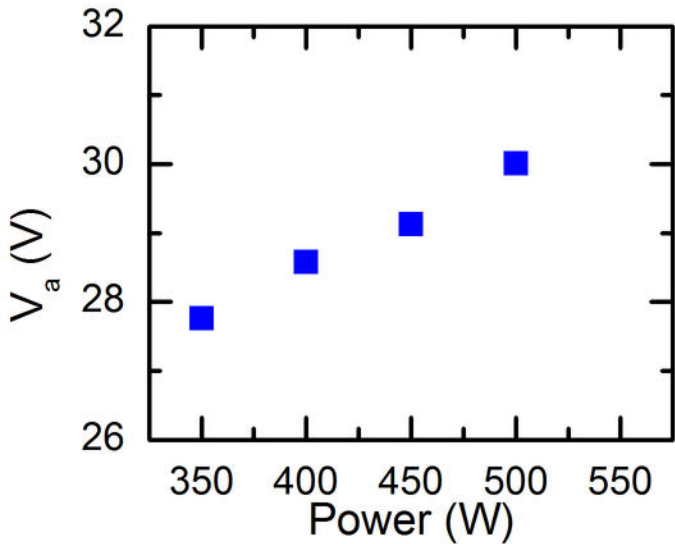


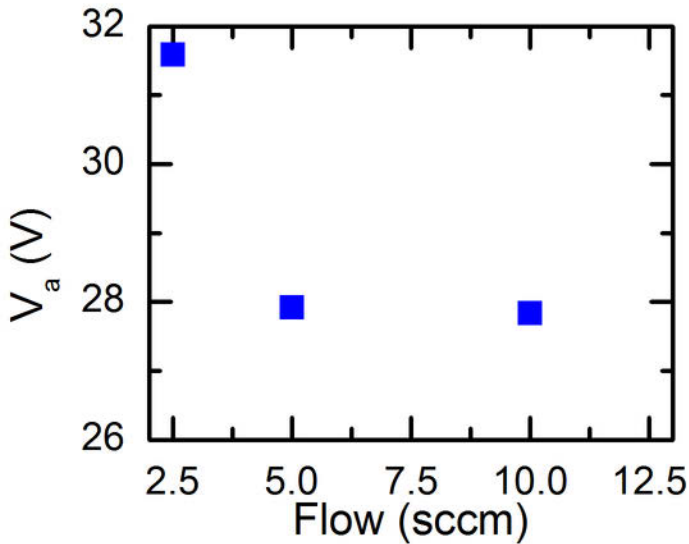


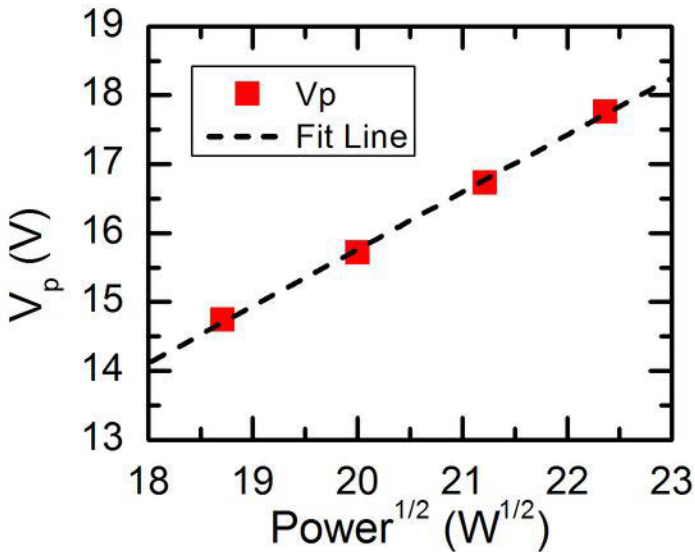


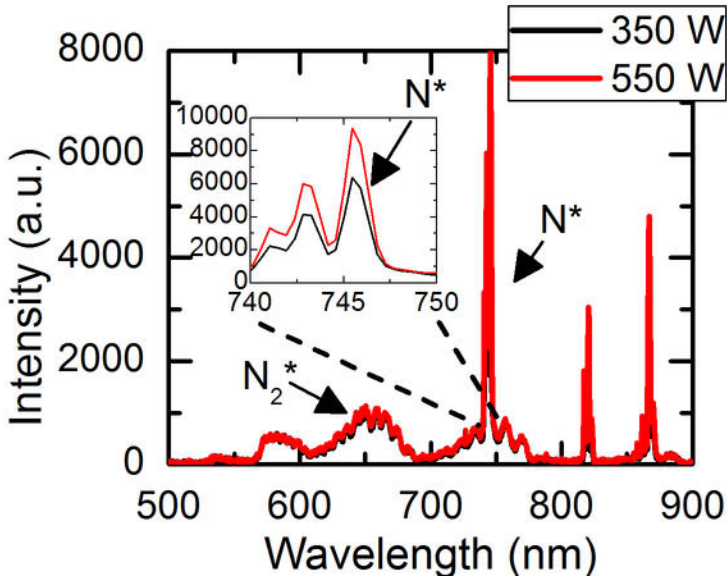


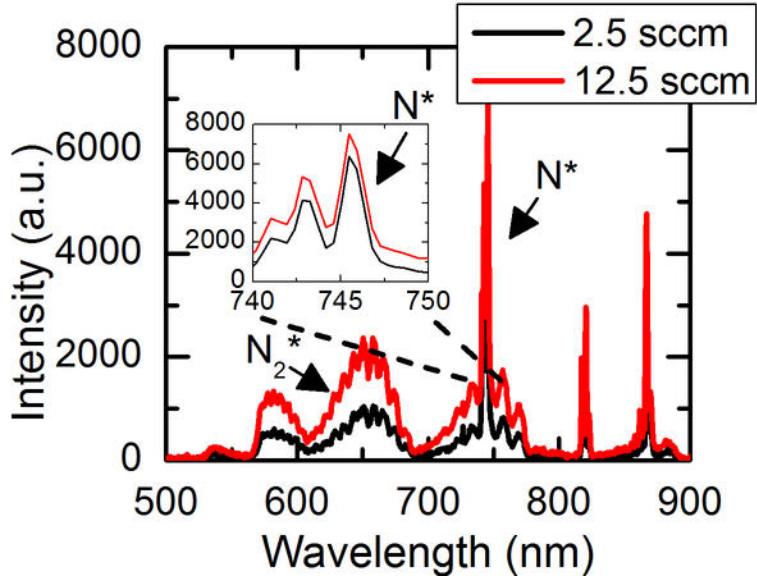


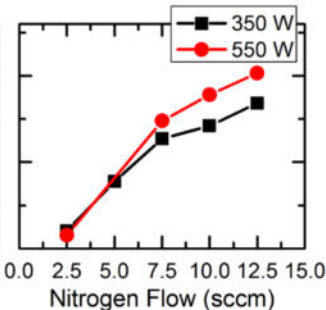
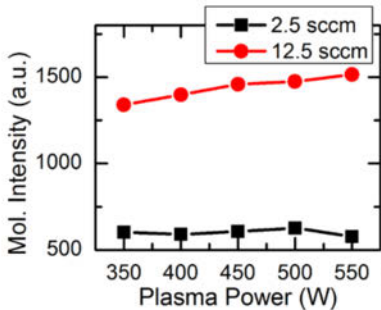
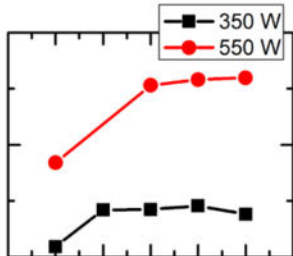
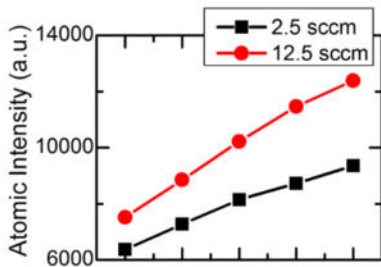


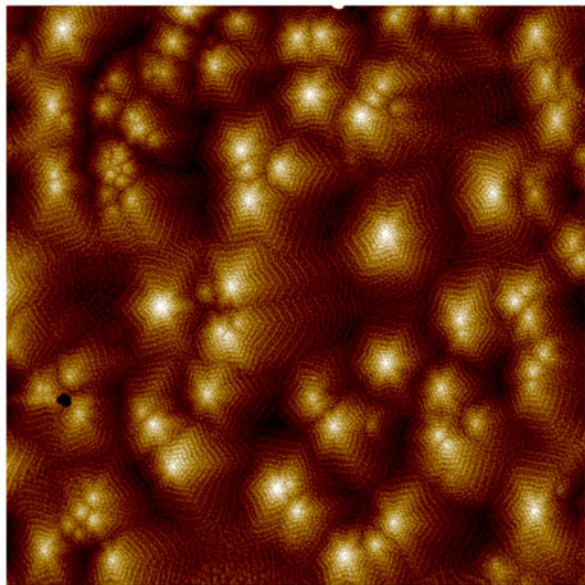










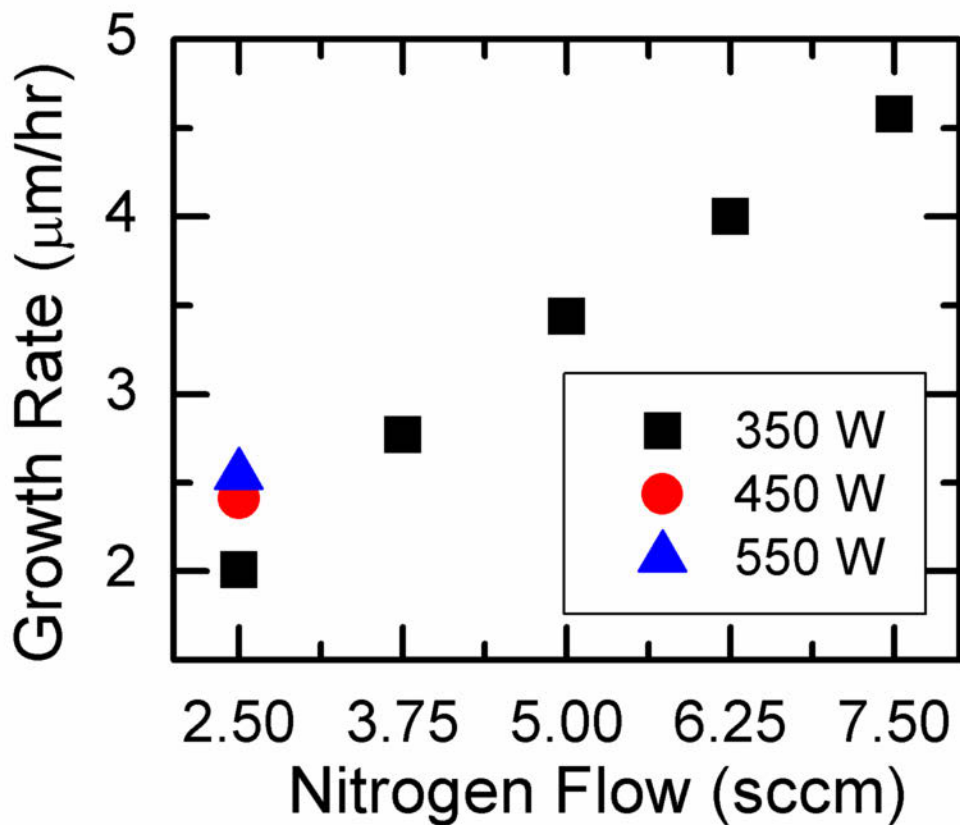


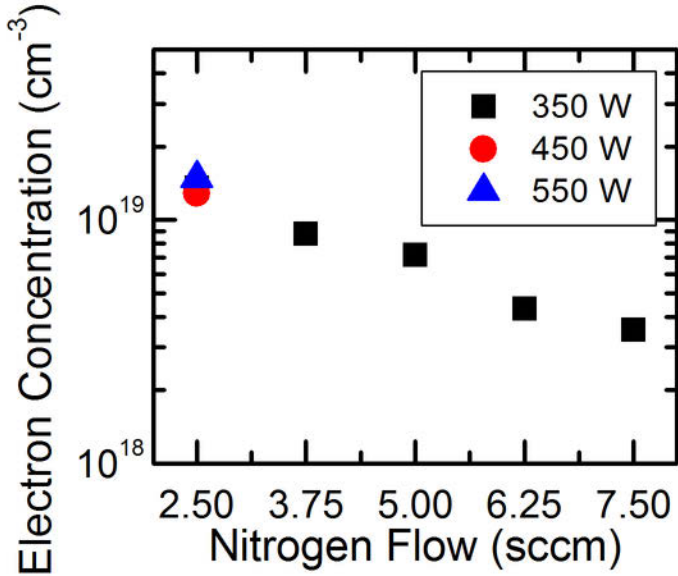
7.8 nm



Height

1.0 μm





$T = 4\text{K}$

Intensity (a.u.)

$1.49 \times 10^{19} \text{ cm}^{-3}$

$1.36 \times 10^{19} \text{ cm}^{-3}$

$1.29 \times 10^{19} \text{ cm}^{-3}$

$8.77 \times 10^{18} \text{ cm}^{-3}$

$7.18 \times 10^{18} \text{ cm}^{-3}$

$4.33 \times 10^{18} \text{ cm}^{-3}$

$3.54 \times 10^{18} \text{ cm}^{-3}$

0.68 0.72 0.76 0.80 0.84

Energy (eV)

